An Explainable Adaptive Temporal Learning Framework for Early Plant Growth Forecasting Using Integrated Soil Environmental and Historical Growth Data

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Abstract— Plant growth is shaped by soil conditions, environmental exposure, and previous development. This paper presents an explainable adaptive temporal learning framework for forecasting future plant height from integrated soil, daily environmental, and historical growth observations. The framework constructs leakage-safe lagged and rolling features; benchmarks persistence, linear regression, Random Forest, XGBoost, GRU, and LSTM; and uses validation-trained stacking to combine tabular and temporal signals. Adaptive feature selection is fitted on the training period only, while SHAP provides local and global explanations for a fitted model. The design is evaluated with chronological partitions using MAE, RMSE, and R squared. A React, Spring Boot, MySQL, and FastAPI implementation separates user authorization, data persistence, and machine-learning inference. The supplied dataset inventory contains environmental and aggregate greenhouse measurements but no repeated plant-height target. Therefore, this paper reports the proposed system and reproducible evaluation protocol without fabricating performance values.

Index Terms— Plant growth forecasting, temporal learning, explainable AI, SHAP, feature selection, precision agriculture.

I. INTRODUCTION

Plant growth monitoring supports evidence-based cultivation, yet manual measurement alone cannot reveal all temporal effects of weather, soil and prior development. Forecasting requires an explicit distinction between measured observations and future estimates. The proposed system forecasts future plant height in centimetres for a stated horizon, and retains the feature snapshot and model version that produced every forecast.

Conventional tabular machine learning can represent nonlinear interactions, but often treats each row independently. Growth trajectories instead reflect accumulated environmental exposure and earlier growth. The proposed framework therefore combines soil properties, daily environmental observations, and a plant-specific historical growth sequence. It is designed as a research decision-support system; it does not diagnose disease, guarantee yield, or convert an attribution into a causal biological claim.

A. Contributions

1) A leakage-safe multimodal formulation for future-height forecasting; 2) comparable tabular and sequential model benchmarks; 3) adaptive feature selection and validation-trained fusion; 4) SHAP explanations linked to the fitted artifact and input snapshot; and 5) an end-to-end web architecture with authenticated data management and auditable prediction persistence.

II. RELATED WORK AND BASE PAPER CONTEXT

Environmental sensing and recurrent modelling have shown that plant growth depends on prior conditions. LSTM-based tomato-height prediction and multimodal greenhouse trajectory modelling motivate sequence-aware forecasts. SHAP has also been used to inspect predictive feature contributions in agricultural applications. However, an accessible, auditable system combining soil data, daily environment, historical height, adaptive selection, model fusion, and time-aware evaluation remains a useful integration problem.

The selected base paper by Condò et al. presents a self-adhesive fiber Bragg grating wearable for physical monitoring of stem elongation. Its goal is measurement through an optical sensor adhered to the plant. In contrast, the present work predicts a later numerical height using prior records and contextual data. The approaches are complementary but are not equivalent: this application does not reproduce, replace, or claim the sensor’s metrological performance.

III. BASE PAPER COMPARISON AND PROPOSED SYSTEM

Aspect	FBG wearable base paper	Proposed forecasting framework
Primary goal	Monitor stem elongation through an attached optical sensor.	Forecast a future plant-height value from prior records.
Core input	FBG wavelength/strain response and hydrogel wearable.	Soil, environmental and timestamped historical growth data.
Output	Physical measurement of elongation.	Predicted future height, model version and SHAP explanation.
Validation	Sensor metrology and in-vivo monitoring.	Chronological ML evaluation: MAE, RMSE, R squared.

Table I. Distinction between the base paper and the proposed project.

A. Working Flow

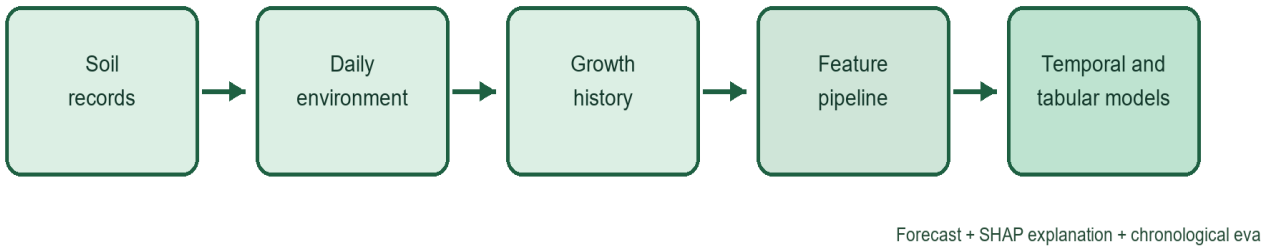


Fig. 1. Proposed workflow from provenance-preserving records to explainable forecast.

At a forecast origin, soil records are joined by plant or location and valid temporal meaning. Daily environmental data are joined by location and timestamp; historical growth is joined by stable plant identity. Lagged height, observed growth rate, rolling environmental summaries, cumulative precipitation, time gaps, and valid soil attributes are then generated. A feature is retained only if it was available at the origin timestamp.

B. Data Integrity and Target Definition

Modality	Example fields	Control
Soil	pH, moisture, nitrogen, phosphorus, potassium, texture	Null is retained for unknown values; source, depth, and date are recorded.
Environment	temperature, humidity, precipitation, solar radiation	Source units are retained; radiation is never renamed as lux.
Growth	plant code, measurement timestamp, plant height	No synthetic daily heights; duplicate timestamps are rejected.

Table II. Data modalities and integrity controls.

The primary target is future plant height: the height measured at the valid observation nearest to origin plus horizon. For irregular sampling, origins without a target within the declared alignment tolerance are excluded. A separate growth-amount target may be trained only as a separately labelled experiment.

IV. MODELS FEATURE SELECTION AND EXPLAINABILITY

A persistence baseline predicts the latest measured height and establishes whether additional modelling has value. Linear regression is a second simple benchmark. Random Forest and XGBoost consume selected tabular features. GRU and LSTM consume chronological sequences grouped by plant so that no sequence crosses plant identities. All transforms, including imputation and scaling, are fitted only on the training period.

Model	Representation	Research role
Persistence	Latest height	Minimum baseline.
Linear regression	Tabular features	Linear benchmark.
Random Forest	Selected tabular features	Nonlinear tree benchmark.
XGBoost	Selected tabular features	Boosted nonlinear benchmark.
GRU / LSTM	Per-plant chronological sequence	Temporal benchmark.
Stacked fusion	Temporally valid base predictions	Integrates tabular and temporal branches.

Table III. Model family and role.

A. Adaptive Feature Selection

The selector is fitted with training data only, using model-based or permutation importance and a defined top-k or threshold rule. Persisted metadata records the selected feature names, their order, preprocessing version, training data version, and model configuration. This makes inference repeatable and prevents test-period information from selecting features.

B. Fusion and SHAP Explanation

The fusion model uses a validation-trained stacking strategy. Base learners are fit on the training period. Temporally valid base predictions form inputs to a meta-regressor, whose configuration is selected on validation data and frozen before final testing. Tree-model SHAP uses a tree explainer. For a neural temporal model, a validated explainer and representative training background are required. For fusion, meta-model contributions must be reported as fusion-level explanations; base-model explanations must not be presented as an explanation of the final fused value.

C. Explanation Output

For every completed forecast, the system records the selected model and version, base output, positive and negative feature contributions, contribution magnitudes, warnings, and input snapshot. SHAP is an explanation of learned model behaviour, not proof that a feature biologically caused growth.

V. IMPLEMENTATION ARCHITECTURE

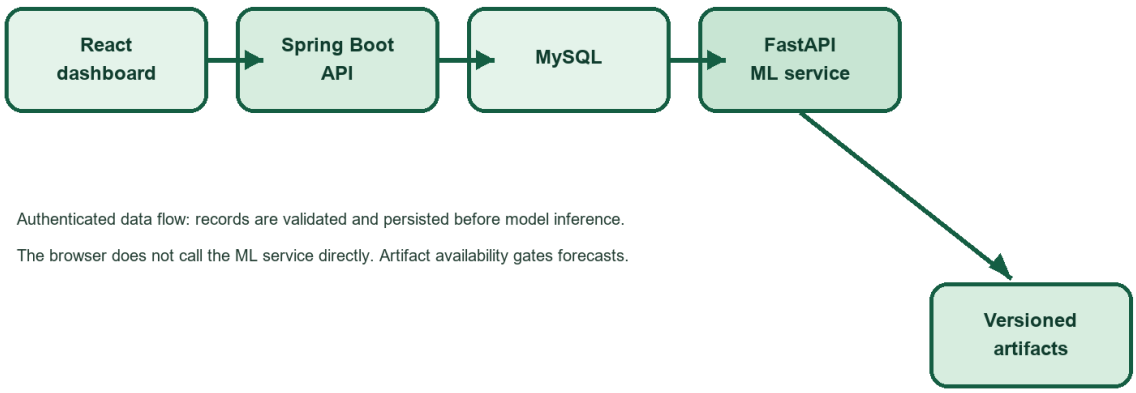


Fig. 2. Full-stack deployment architecture and separation of responsibilities.

The React client retrieves data only from Spring Boot. Spring Security verifies JWTs and ownership before plants, observations, or predictions are read. The backend then gathers validated records, calls the FastAPI service, persists the returned model metadata and explanation, and returns a frontend-friendly response. A missing or incompatible artifact produces an explicit error rather than a fallback prediction.

VI. EVALUATION PROTOCOL AND RESULTS REPORTING

Forecasting evaluation must be chronological rather than randomly shuffled. The earliest observations are reserved for training, the following observations for model selection, and the newest observations for the final held-out test. Model comparison is valid only when all compared models use the same target, horizon, and test observations.

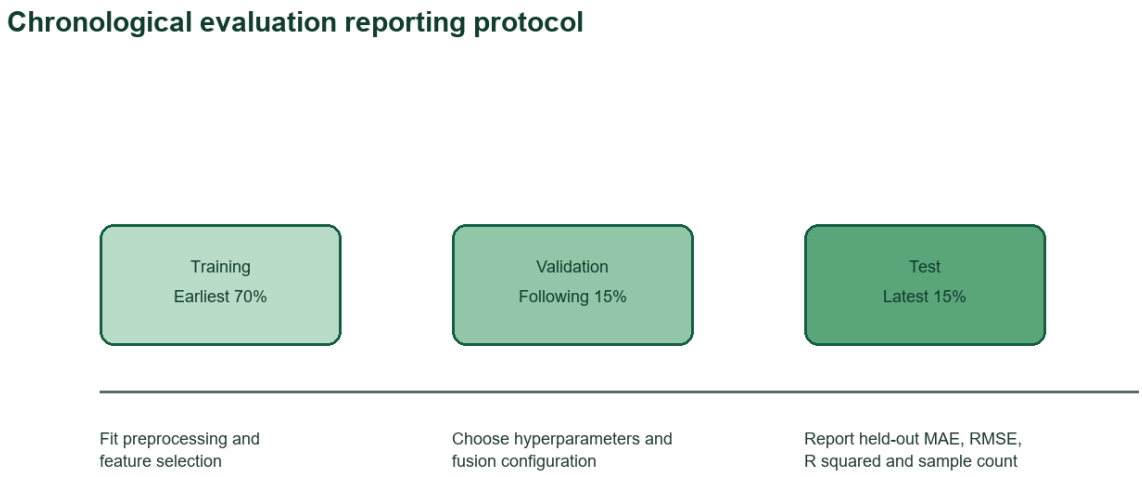


Fig. 3. Chronological evaluation protocol. No performance bars are shown until valid experimental results exist.

Metric	Definition	Interpretation
MAE	mean absolute error	Average absolute future-height error in centimetres.
RMSE	square root of mean squared error	Penalizes larger height errors more strongly.
R squared	1 - residual sum of squares / total sum of squares	Explained variation; guarded for small or constant sets.

Table IV. Regression measures to be reported from held-out predictions.

A. Current Dataset Observation

The supplied aggregate greenhouse dataset includes weather and fruit/biomass measures but marks plant_height_cm as not provided. It therefore cannot validly train, tune, or score a plant-height forecaster. This is a data availability observation, not an empirical finding about model quality. The implementation must receive repeated timestamped measurements per plant before a seven-day height experiment can be claimed.

B. Results Reporting Plan

When a compatible dataset becomes available, Table V will be filled from the pipeline output with model version, target type, horizon, test date range, sample count, MAE, RMSE, and R squared. The bar-chart form commonly used in conference papers will then be generated from those stored values. It is intentionally not prefilled here: fabricated metrics would invalidate the research contribution.

Model	Horizon	MAE cm	RMSE cm	R squared	Status
Persistence	7 days	Not evaluated	Not evaluated	Not evaluated	Awaiting height data
Random Forest	7 days	Not evaluated	Not evaluated	Not evaluated	Awaiting height data
XGBoost	7 days	Not evaluated	Not evaluated	Not evaluated	Awaiting height data
GRU / LSTM / Fusion	7 days	Not evaluated	Not evaluated	Not evaluated	Awaiting sequential height data

Table V. Honest evaluation status at the time of writing.

VII. APPLICATION WORKFLOW AND OBSERVATION MANAGEMENT

A user registers a plant and its growing location, then enters or imports soil records, environmental observations, and historical growth measurements. Each input retains a source, timestamp, quality status, and raw value. CSV import validation reports accepted rows, rejected rows, missing-value summary, duplicate-record summary, and row-level error information rather than silently discarding data.

Step	System activity	Audit artefact
1	Authenticate user and enforce ownership.	JWT identity and authorized resource scope.
2	Register plant and location.	Stable plant code and location reference.
3	Store validated observations.	Timestamp, source, quality status.
4	Request supported forecast.	Feature snapshot and artifact version.
5	Persist model result and SHAP explanation.	Prediction timestamp and explanation JSON.
6	Record future actual height.	Absolute error and evaluation history.

Table VI. End-to-end operational flow.

A. Data Quality Rules

The request is rejected for unknown plant identifiers, negative heights, invalid pH, humidity outside 0 to 100 percent, duplicate growth timestamps, unsupported horizons, insufficient historical sequence length, or unavailable artifacts. Missing values are never replaced with zero. Environmental gaps are represented as gaps and are not forward-filled unless a documented, training-derived policy justifies it.

B. Operational Outputs

The dashboard distinguishes measured records from forecasts, shows “Not available” for absent values, and labels every forecast as a prediction rather than a measurement. The results page reports the current height, forecast horizon, predicted future height or growth amount, increase relative to current height where appropriate, model version, data-quality warnings, and actual-versus-predicted error after the future measurement is recorded.

C. Responsible Interpretation

A plotted prediction is only a model estimate. It must not be described as improved crop yield, plant diagnosis, or a guaranteed outcome. The model comparison becomes a research result only after the chronological held-out experiment has completed successfully. The current system design makes this distinction visible in both the API and user interface.

VIII. CONCLUSION AND FUTURE WORK

This work defines a full-stack, explainable temporal framework for forecasting future plant height from soil, environmental, and historical observations. Its central contribution is not merely a model list but a reproducible pathway from validated records to leakage-safe training, artifact-gated inference, auditable explanations, and chronological evaluation. The immediate research priority is collection or licensing of repeated timestamped plant-height measurements at a sampling frequency that supports the intended horizon. Future work will evaluate uncertainty, species transfer, location transfer, and the potential integration of sensor measurements as an additional input stream.

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APPENDIX: REPRODUCIBILITY CHECKLIST

Item	Requirement
Target	Future plant height in cm, with a declared horizon and alignment tolerance.
Split	Chronological train, validation, and final held-out test periods.
Transforms	Fit imputation, scaling, encoding, and feature selection on training only.
Artifacts	Store feature order, pipeline version, model version, configuration, and evaluation details.
Reporting	Give sample count, date range, MAE, RMSE, R squared, and data limitations.
Explainability	Link the explanation to the exact model and transformed input; do not imply causality.

Table VII. Minimum reproducibility requirements for the final experiment.

Author information note: E-mail addresses are shown only where supplied by the project team. This six-page draft uses a standard conference-paper layout and should be adjusted to the final conference’s official template before submission.